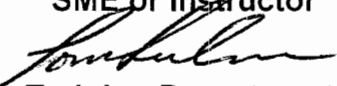
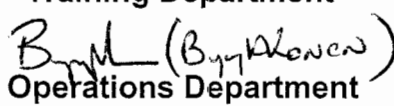


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

STATION:	SALEM		
SYSTEM:	ECCS		
TASK:	Perform forward Flow IST of ECCS Accumulator Check Valves		
TASK NUMBER:	N0060160201		
JPM NUMBER:	11-01 ILOT Sim b		
ALTERNATE PATH:	☒	K/A NUMBER:	006 A4.02
APPLICABILITY:		IMPORTANCE FACTOR:	
	EO ☐		4.0
	RO ☒		RO
	STA ☐		3.8
	SRO ☒		SRO
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-ST.SJ-0006, Inservice Testing Safety Injection Valves Mode 6, Rev. 13		
TOOLS AND EQUIPMENT:	Stopwatch		
VALIDATED JPM COMPLETION TIME:	15 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	06-04-12
Validated By:	Mulford SME or Instructor	Date:	08-23-12
Approved By:	 Training Department	Date:	9/13/12
Approved By:	 Operations Department	Date:	9/12/12
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

NAME: _____

DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

TASK NUMBER: N0060160201

SIMULATOR SETUP

IC-242 This IC was developed by: Simulating Mode 6 conditions. Depressurizing RCS with RHR remaining in service. Opening both PORVs. Simulating cavity level 126' by filling PZR full.

Remote: SJ01D 21SJ54 breaker 2AY2EP2D

Remote: SJ02D 22SJ54 breaker 2BY2EP2D

ET-1 ka103anl 22SJ54 open PB. Tied to:

22 Accumulator Channel I pressure (**I/O A105**) Final Value 33 with a ramp time of 38 seconds.

22 Accumulator Channel II pressure (**I/O A106**) Final Value 33 with a ramp time of 37 seconds.

INITIAL CONDITIONS:

Unit 2 is in MODE 6 with the Rx vessel head removed.

The control room is preparing to perform ECCS Accumulator Check Valve testing.

A second control room operator is available.

A NEO is stationed in the field for any operations required.

Reactor cavity level is 126'.

Refueling Transfer Tube Gate Valve is shut.

INITIATING CUE: Perform forward flow testing of 21-24 ECCS Accumulators Check Valves IAW S2.OP-ST.SJ-0006. This is the normally scheduled surveillance.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Perform forward flow test of 21 ECCS Accumulator and determine it is SAT.
2. Perform forward flow test of 22 ECCS Accumulator and determine it is UNSAT.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide copy of S2.OP-ST.SJ-0006 with prerequisites complete and calibration data filled in.	Cue if required: If asked during JPM performance about cavity level, evaluator will respond that the normal control room staff is monitoring level, and it remains stable at 126'.		
	3.0	PRECAUTIONS AND LIMITATIONS.	Reviews Precautions and Limitations		
	5.1.1	RECORD 21 Accumulator Initial Data in Attachment 2, Section 1.0.	Records 21 Accumulator Initial Data in Attachment 2, Section 1.0. as 69.2 psig and 96.6 %.		
*	5.1.2	CLOSE 2AY2EP2D, 21SJ54 ACCUMULATOR OUTLET VALVE.	Contacts NEO and directs them to shut 2AY2EP2D, 21SJ54 ACCUMULATOR OUTLET VALVE. Simulator Operator: Modify Remote SJ01D from tagged to untagged to shut 2AY2EP2D and report back to control room.		
	5.1.3	At Panel 2RP4, ENSURE 21SJ54 in the VALVE OPERABLE position.	At Panel 2RP4, checks 21SJ54 is in the VALVE OPERABLE position.		
	5.1.4	IF Scheduled Surveillance testing is to be performed (Refer to P&L Step 3.6), THEN:			
*	5.1.4.A	A. Simultaneously PERFORM the following: OPEN 21SJ54, 21 ACCUM OUT START the Stopwatch	Simultaneously depresses the open PB for 21SJ54 and starts stopwatch.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.1.4.B	B. When 21 Accumulator pressure is observed to be 35 psig, Simultaneously PERFORM the following: • STOP the Stopwatch • CLOSE 21SJ54	Simultaneously depresses the 21SJ54 close PB and stops the stopwatch when 21 Accumulator pressure indicates 35 psig.		
* *	5.1.4.C	PERFORM the following IAW Attachment 2, Section 2.0: • RECORD Stopwatch Number and Reading (A) in Attachment 2, Section 2.0. • CALCULATE Accumulator Blowdown Time (C) by subtracting Disk Movement Time (B) from Stopwatch Reading (A) AND RECORD in Attachment 2, Section 2.0. • DIRECT a second Operator to perform Independent Verification of the calculation performed IAW Attachment 2, Section 2.0.	Records stopwatch number and reading (A) in Attachment 2 Section 2.0 Calculates Accumulator Blowdown Time (C) by subtracting Disk Movement Time (B-1.15 seconds) from Stopwatch Reading (A) AND records in Attachment 2, Section 2.0. Directs a second Operator to perform Independent Verification of the calculation performed IAW Attachment 2, Section 2.0. Cue: IV is complete SAT.		
*	5.1.4.D	RECORD 21SJ55 and 21SJ56 OPEN "Test Results" by initialing the SAT or UNSAT column using the Acceptance Criteria in Attachment 2, Section 2.0.	Records 21SJ55 and 21SJ56 OPEN "Test Results" by initialing the SAT column using the Acceptance Criteria in Attachment 2, Section 2.0.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.1.5	IF Post-Maintenance Operability (Valve Exercise) testing is to be performed * (Refer to P&L Step 3.6), * THEN: A. OPEN 21SJ54, 21 ACCUM OUT. B. When 21 Accumulator level has decreased by at least 20%, CLOSE 21SJ54, 21 ACCUM OUT. C. RECORD 21SJ55 and 21SJ56 VALVE EXERCISE "Test Results" by initialing the SAT or UNSAT column using the Acceptance Criteria in Attachment 2, Section 3.0.	Determines Post Maintenance Operability testing is N/A.		
	5.1.6	OPEN 2AY2EP2D, 21SJ54 ACCUMULATOR OUTLET VALVE.	Contacts NEO and directs them to open 2AY2EP2D, 21SJ54 ACCUMULATOR OUTLET VALVE. Simulator Operator: Modify REMOTE SJ01D to tagged to open 2AY2EP2D.		
	5.1.7	DIRECT a second Operator to perform Restoration Independent Verification of the components identified in Attachment 6, Section 1.0.	Cue: Restoration Independent Verification of the components identified in Attachment 6, Section 1.0. is complete SAT.		
	5.2.1	RECORD 22 Accumulator Initial Data in Attachment 3, Section 1.0.	Records 22 Accumulator Initial Data in Attachment 3, Section 1.0. as 67.4 psig and 97.6 %.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.2.2	CLOSE 2BY2EP2D, 22SJ54 ACCUMULATOR OUTLET VALVE.	Contacts NEO and directs them to shut 2BY2EP2D, 22SJ54 ACCUMULATOR OUTLET VALVE. Simulator Operator: Modify Remote SJ02D from tagged to untagged to shut 2BY2EP2D.		
	5.2.3	At Panel 2RP4, ENSURE 22SJ54 in the VALVE OPERABLE position.	At Panel 2RP4, checks 22SJ54 is in the VALVE OPERABLE position.		
	5.2.4	IF Scheduled Surveillance testing is to be performed (Refer to P&L Step 3.6), THEN:			
*	5.2.4.A	A. Simultaneously PERFORM the following: • OPEN 22SJ54, 22 ACCUM OUT • START the Stopwatch	Simultaneously depresses the open PB for 22SJ54 and starts stopwatch. Simulator Operator: Ensure ET-1 is TRUE when 22SJ54 Open PB is depressed.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.2.4.B	B. When 22 Accumulator pressure is observed to be 35 psig, Simultaneously PERFORM the following: • STOP the Stopwatch • CLOSE 22SJ54	Simultaneously depresses the 22SJ54 close PB and stops the stopwatch when 22 Accumulator pressure indicates 35 psig. Simulator Operator: If 22SJ54 close PB is <u>not</u> depressed, DELETE overrides A105 and 106 when they reach their full ramp in time.		
* *	5.2.4.C	PERFORM the following IAW Attachment 3, Section 2.0: • RECORD Stopwatch Number and Reading (A) in Attachment 3, Section 2.0. • CALCULATE Accumulator Blowdown Time (C) by subtracting Disk Movement Time (B) from Stopwatch Reading (A) AND RECORD in Attachment 3, Section 2.0. • DIRECT a second Operator to perform Independent Verification of the calculation performed IAW Attachment 3, Section 2.0.	Records stopwatch number and reading (A) in Attachment 3 Section 2.0 Calculates Accumulator Blowdown Time (C) by subtracting Disk Movement Time (B-1.15 seconds) from Stopwatch Reading (A) AND records in Attachment 3, Section 2.0. Directs a second Operator to perform Independent Verification of the calculation performed IAW Attachment 3, Section 2.0. Cue: IV is complete SAT.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.2.4.D	RECORD 22SJ55 and 22SJ56 OPEN "Test Results" by initialing the SAT or UNSAT column using the Acceptance Criteria in Attachment 3, Section 2.0.	Records 22SJ55 and 22SJ56 OPEN "Test Results" by initialing the UNSAT column using the Acceptance Criteria in Attachment 3, Section 2.0. based on blowdown time >27.79 seconds.		
	5.2.5	IF Post-Maintenance Operability (Valve Exercise) testing is to be performed * (Refer to P&L Step 3.6), * THEN: A. OPEN 22SJ54, 21 ACCUM OUT. B. When 22 Accumulator level has decreased by at least 20%, CLOSE 22SJ54, 22 ACCUM OUT. C. RECORD 22SJ55 and 22SJ56 VALVE EXERCISE "Test Results" by initialing the SAT or UNSAT column using the Acceptance Criteria in Attachment 3, Section 3.0.	Determines Post Maintenance Operability testing is N/A.		
	5.2.6	OPEN 2BY2EP2D, 22SJ54 ACCUMULATOR OUTLET VALVE.	Contacts NEO and directs them to open 2BY2EP2D, 22SJ54 ACCUMULATOR OUTLET VALVE. Simulator Operator: Modify REMOTE SJ02D to tagged to open 2BY2EP2D.		

JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- M 1. Task description and number, JPM description and number are identified.
- M 2. Knowledge and Abilities (K/A) references are included.
- M 3. Performance location specified. (in-plant, control room, or simulator)
- M 4. Initial setup conditions are identified.
- M 5. Initiating and terminating Cues are properly identified.
- M 6. Task standards identified and verified by SME review.
- M 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- M 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 13 Date 8/23/12
- M 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- N/A 10. ☒ If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

N. Mulford

Date: 8/23/12

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Perform forward Flow IST of ECCS Accumulator Check Valves

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.2.7	DIRECT a second Operator to perform Restoration Independent Verification of the components identified in Attachment 6, Section 2.0.	Cue: Restoration Independent Verification of the components identified in Attachment 6, Section 2.0. is complete SAT.		
			Cue: JPM is complete.		

JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

Unit 2 is in MODE 6 with the Rx vessel head removed.
The control room is preparing to perform ECCS Accumulator Check Valve testing.
A second control room operator is available.
A NEO is stationed in the field for any operations required.
Reactor cavity level is 126'.
Refueling Transfer Tube Gate Valve is shut.

INITIATING CUE:

Perform forward flow testing of 21-24 ECCS Accumulators Check Valves IAW S2.OP-ST.SJ-0006.
This is the normally scheduled surveillance.